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Using ecological forest site mapping for long-term habitat suitability assessments in wildlife conservation—Demonstrated for capercaillie (*Tetrao urogallus*)

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ABSTRACT

Although key factors for vegetation composition and structure, site and soil condition have received little attention as predictors of habitat suitability in wildlife ecology to date. Using the example of capercaillie (*Tetrao urogallus*), an indicator species for open, well-structured forest habitats, we evaluated the potential use of ecological forest site mapping for the identification of areas where the preferred vegetation structures are supported by the prevailing soil conditions. These are sites that we, therefore, expected to be of long-term relevance to the species.

Employing an expert model, we evaluated data from the state's forest site and soil condition database including soil texture, soil type, humus type, nutrient status and hydrological regime. The data were aggregated to a soil condition index (I_s), quantifying the potential to support capercaillie-relevant vegetation types. First, we assessed the correlation of I_s with the current vegetation conditions in the study area. Subsequently, we evaluated the short-term (15 years) and long-term (100 years) changes to the capercaillie distribution with respect to I_s at different spatial scales and calculated a threshold for the prediction of long-term capercaillie presence.

Although the vegetation structure in the study area was largely influenced by forestry, I_s was correlated with all investigated vegetation structure variables that determine habitat suitability for capercaillie. Soil condition proved to be an important indirect predictor of long-term capercaillie inhabitation, performing best when considering an area exceeding the size of a large capercaillie home range. The long-term range contractions corresponded significantly with the spatial pattern of suitable soil conditions, the short-term development demonstrated a similar trend. The results indicated an ongoing retreat from degenerating secondary habitats, created by historic land use techniques on sites with low I_s, to sites with high I_s values, which should be considered when defining priority areas for conservation. Forest ecological site mapping provides detailed, area-wide data that allow for a spatially explicit evaluation of the vegetation development. Therefore, it can facilitate the integration of conservation targets in forest management, for species dependent upon specific vegetation conditions.

1. Introduction

Soil condition, in addition to site characteristics such as climate and water supply, is one of the main predictors of vegetation composition and vegetation structure. It is for this reason that site related data have been collected for large areas across Europe since the middle of the 20th century. This process began with the forestry site classification (Krauss et al., 1949; Schlenker, 1950; Aldinger et al., 1998), followed by site mapping for agriculture and landscape development (e.g., Ellenberg, 1951). In ecological forest site mapping, factors relevant for tree growth and vegetation development such as soil parameters (soil texture, soil type, humus form, soil nutrient and soil moisture regime) and data on local climate and topography are assessed area wide at a high resolution (scale 1:10,000). The data are used for silvicultural planning (e.g., Krauss et al., 1949; Schlenker, 1950; Englisch et al., 2001), primarily in order to optimise management targets such as timber production. Subsequently, with the trend towards sustainable forestry, ecological aspects such as the potential natural vegetation grew in importance.

In view of the limited resources available for conservation management on the one hand, and the increasing demands on nature conservation due to national and international legislation (e.g., Natura 2000) on the other, the potential use of existing

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inventories or survey data for conservation management has become an important issue. The ongoing trend towards multifunctional forestry, with improving biodiversity in managed forests among the major objectives, requires solutions for the integration of conservation aspects into forestry practice. Taking into consideration the development potential of sites, and focusing on those areas where the prevailing ecogeographic features naturally support conditions suitable for specific target species can be a crucial issue for sustainable conservation planning. As yet, this approach is not commonly applied in the area of animal conservation. Although many wildlife species are closely linked to specific vegetation conditions, site and soil conditions have received little attention as indirect predictors of habitat suitability so far. Whereas climate and topographic variables are often included in habitat models, soil conditions are rarely integrated, usually only when a species is directly affected (Kolström and Lumatjärvi, 1999; Courtright et al., 2001, but refer to Wertz, 1966; Moores et al., 1996; Manning, 2000). In most studies assessing habitat quality, vegetation conditions are mapped directly and integrated into habitat suitability models (e.g., Luoto et al., 2002; McComb et al., 2002; Storch, 2002). These direct or resource variables (sensu Austin, 1980, 1985; Austin and Smith, 1989) are preferred to surrogate variables (that impact the species indirectly by determining the vegetation) for a number of reasons (Guisan et al., 1999; Van Horne, 2002; Guisan and Thuiller, 2005). First, they are often more precise at a local scale, and secondly, they are directly perceived or consumed by the species and, therefore, relate to functional processes of habitat selection (e.g., Austin, 2002; Heglund, 2002; Guisan and Thuiller, 2005). The focus on vegetation condition is advantageous where the objective is to model the current habitat suitability. However, problems arise when seeking to determine the long-term potential of a landscape to provide suitable habitat for a particular species (e.g., Braunisch and Suchant, 2007).

In the cultivated landscapes of central Europe, the natural vegetation has frequently been profoundly influenced by human land use (Brückner, 1984; Jahn et al., 1990; Stuber and Bürgi, 2001). Historical forms of land use such as intensive pasturing, litter raking and coppicing, for example, led to secondary vegetation forms and associations, forming temporary habitats for various animal species (e.g., Reif, 1998; Greatorex-Davies and Marrs, 1992; Guido et al., 1999; Chemini and Rizzoli, 2003). After the cessation of management, however, these habitats began to regress, subject to the natural starting conditions, mainly geology, soils and climate (Glatzel, 1991). The consideration of these conditions may allow for the prediction of patterns in vegetation structure development and their integration into conservation planning. As site conditions are less susceptible to anthropogenic influences and management regimes than the vegetation itself, site and soil inventory data may be useful for assessing areas with a long-term potential for providing suitable habitat conditions, particularly for species that are indicative of specific vegetation types and structures.

Capercaillie (*Tetrao urogallus*), a large European forest grouse, is an indicator of well-structured boreal and montane forest habitats characterised by an intermediate canopy cover and an abundant ground vegetation (Scherzinger, 1989, 1991; Boag and Rolstad, 1991; Storch, 1993a,b, 1995; Cas and Adamic, 1998; Suter et al., 2002), ideally dominated by bilberry (*Vaccinium myrtillus*) (e.g., Storch, 1993a,b, 1995; Selås, 2000, 2001). Whereas the boreal forest belt is continuously inhabited by capercaillie, the mostly small and declining populations in central Europe are limited to higher altitudes in mountain regions (Klaus et al., 1989; Storch, 2000). The Black Forest mountain range hosts the second largest capercaillie population in central Europe (Braunisch and Suchant, 2006; Storch, 2000). Up until the end of the 19th century capercaillie inhabited forest areas down into the lowlands (Suchant, 2002). At this time, many central European forests, including the Black Forest, were not only sources of timber and fire wood but were also intensively used as forest pasture, for litter raking and pollarding, which created open forest structures (Brückner, 1984; Schmidt, 1989, 2005; Jahn et al., 1990). Moreover, the historic use of biomass components other than wood led to a severe depletion of nutrients from the topsoil, and impacted greatly on their acid neutralising capacity (Glatzel, 1991; Prietzel and Kaiser, 2005). The consequential decline in tree growth, the altered tree species composition, and the development of acidophilic ground vegetation like bilberry (Sayer, 2006) also facilitated the range expansion of several bird species like capercaillie with preferences for open, well-structured forests (Suchant, 2002; Gatter, 2004). With the change over to sustainable, long-rotation forestry, the nutritional status of the devastated soils slowly recovered, brought about by an accumulation of organic carbon and nitrogen (Mellert et al., 2004; Prietzel et al., 2006). This process, locally furthered by active fertilisation and atmospheric nitrogen deposition (Bürger-Arndt, 1994), was followed by changes in vegetation structure and a degeneration of the secondary habitats.

Over the last hundred years, a continual decline in the central European capercaillie populations has been observed, along with a strong reduction of the species' distribution (Klaus et al., 1989; Storch, 2000; for the Black Forest specifically refer to Suchant and Braunisch, 2004; Braunisch and Suchant, 2006). Habitat deterioration is considered to be a main cause, however, it is mainly attributed to active habitat changes brought about by forestry practices, such as the plantation of even-aged stands, the use of fast-growing conifers, or the recent move towards close-to-nature forestry leading to increased stocking densities (e.g., Schröder, 1974; Rolstad and Wegge, 1989; Sjöberg, 1996; Cas and Adamic, 1998).

We hypothesise that the spatial pattern of the contraction of the capercaillie range reflects at least partly a retreat of the species from degenerating secondary habitats to sites where the prevailing site conditions – with soil conditions assumed to be a main factor – naturally support suitable vegetation structures. To test this hypothesis, we grouped different soil condition parameters in a soil condition index quantifying a site's potential to support the vegetation conditions preferred by capercaillie. The index was employed to answer the following questions:

(1) Do soil conditions relate to capercaillie-relevant vegetation structures and local habitat suitability in managed forest ecosystems, where the natural vegetation structures are largely overridden by forestry?
(2) Does the spatial pattern of the contraction of the capercaillie range over the last 100 years coincide with the spatial pattern of suitable soil conditions, and is this process still ongoing?
(3) Can the soil condition index be used to identify sites where long-term capercaillie inhabitation is likely and habitat restoration measures can thus be expected to be most effective?

Apart from quantifying the importance of soil conditions for capercaillie habitats, we demonstrate the practical application of ecological forest site and soil inventories in wildlife conservation and management.

2. Methods

2.1. Study area

The study area comprises the Black Forest, a mountain range in south western Germany, defined by the growth areas *Schwarzwald*

Fig. 1. Black Forest study area. The capercaille distribution (monitoring period 1998–2003, indicated in dark grey) covers about 51,000 ha. The locations of the lekking sites from 1902 are illustrated – white dots: still inhabited; grey dots: abandoned sites. A smaller pilot area for vegetation mapping was delineated in the south western part of the study area.

and *Baar-Wutach* (Aldinger et al., 1998) and covering approximately 7000 km^2 (Fig. 1). Two thirds of the total area are forested, with Norway spruce (*Picea abies*, 49%), European silver fir (*Abies alba*, 19%) and beech (*Fagus sylvatica*, 12%) being the dominant tree species. The geology of the southern part of the study area is mainly determined by the older granite and gneiss basement and in the northern and eastern parts by a younger overlying layer of bunter sandstone. The macroclimate is strongly influenced by the altitudinal gradient, ranging from 120 to 1493 m a.s.l., with an average annual temperature ranging between 4 °C at the higher elevations and 10.4 °C in the lowlands (1st and 9th quantile, according to Gauer and Aldinger, 2005). Capercaillie is currently present on about 51,000 ha in the higher regions, with 258 cocks counted in 2003 (Braunisch and Suchant, 2006). The correlation between soil condition and forest vegetation structure was evaluated in a pilot area of 98 km^2 located in the southern part of the study area (Fig. 1). The altitude ranged between 466 and 1493 m a.s.l.

2.2. Environmental data

2.2.1. Soil condition data obtained from ecological forest site mapping (study area)

The soil condition data used in this study were obtained from the forest site and soil condition database of the federal state Baden-Wuerttemberg, which contains area-wide data for the entire public forest (state and communal forests) and a significant proportion of the private forest in the Black Forest study area (Aldinger et al., 1996; Höhne, 1996). The soil condition parameters were mapped on the forest stand level (spatial scale 1:10,000) in a 'dual approach' according to the method described by Krauss et al. (1949), Schlenker (1950, 1964) and Mühlhäusser et al. (1983):

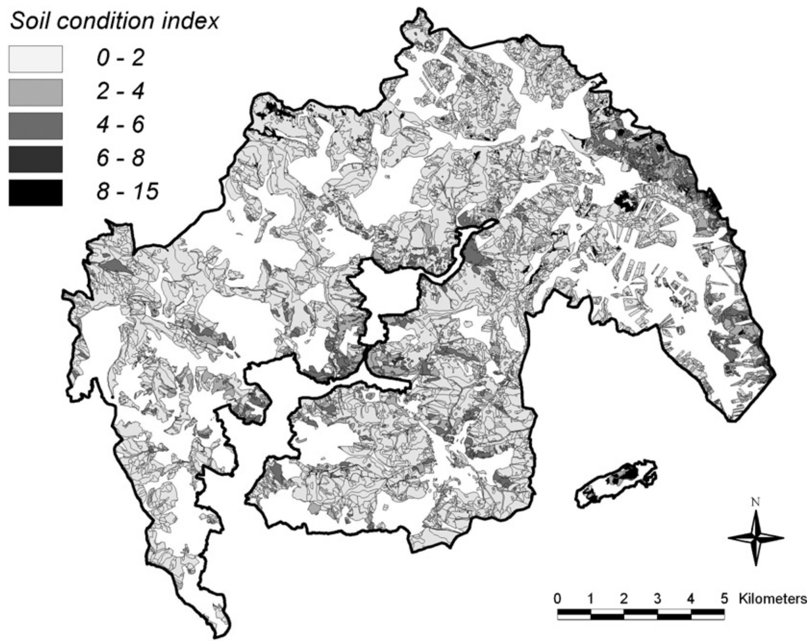

Fig. 2. The pilot area with its units of soil condition delineated by an ecological forest site and soil condition mapping.

First, sites are coarsely demarcated based on characteristics of local climate and topography. Secondly, soil conditions are examined in the field using soil drillings taken on a 30 m × 30 m sample grid, locally reduced to 20 m × 20 m on very heterogeneous sites. Based on the results homogeneously structured site units are delineated that exhibit similar conditions for vegetation growth and can be readily used for forest planning. The site mapping units for the pilot area are presented in Fig. 2.

We evaluated the soil parameters humus type (HTY), soil type (STY), soil texture (STX), nutrient status (NS) and hydrological regime (HR) for $N = 130{,}514$ mapping units. The average size of a unit was 2 ha (min.: 0.01 ha, max.: 357 ha) in the study area. To facilitate the intersection of the soil condition parameters with data mapped for different spatial units, we converted the soil maps to a 30 m × 30 m grid, which approximated the original soil sample resolution.

2.2.2. Vegetation structure data (pilot area)

We assessed capercaillie-relevant vegetation structure variables for the forested part of the pilot area (98 km^2) as described by Suchant (2002). Forest type, canopy cover, successional stage, the number of gaps, ground vegetation and bilberry (*Vaccinium myrtillus*) cover were mapped in the field for forest stand units ($N = 2211$). The mean size of a forest stand in the pilot area was 4 ha (min.: 1 ha, max.: 50 ha).

2.3. Capercaillie data

Capercaillie has been continuously monitored in the Black Forest since 1988. Capercaillie distribution is mapped every 5 years, based on all direct and indirect evidence of capercaillie presence provided by foresters, hunters, ornithologists and the authors' own field work. Areas classified as 'inhabited' (Fig. 1) are delineated using the minimum polygon that included all capercaillie records with a maximum distance of 1000 m proximity to each other collected within the 5 year period (cf. Braunisch and Suchant, 2006).

To analyse the pattern of the short-term development (1988–2003) of the capercaillie distribution, we evaluated the distribution maps produced for the three previous monitoring periods (1988–1993, 1993–1998, 1998–2003). We distinguished between the status of areas that were '*continuously inhabited*', '*newly populated*' or '*abandoned*' within the last 15 years, as defined in Table 1. Areas classed as '*fluctuation*' (Table 1) and areas that were mapped as newly populated or abandoned but fell below a minimum size of 50 ha were excluded from the analysis.

The historical capercaillie distribution was adopted from a complete survey of lekking sites in the eastern part of the Black Forest study area. The survey was conducted by the gamekeepers of the Fürstenberg principality in 1902 (Suchant, 2002) and provided the exact coordinates for 156 capercaillie leks (Fig. 1). In order to assess the long-term development of the capercaillie distribution (1902–2003) we compared two cases: the areas surrounding historical lekking sites which are '*still inhabited*' by capercaillie and those now '*abandoned*'.

2.4. Soil condition index

2.4.1. Expert model

We employed an expert model for the gradated evaluation of soil parameters and their combination in a soil condition index.

Table 1
Population status of areas with capercaillie occurrence within the last 15 years, as defined by Braunisch and Suchant (2006)

Status of an area	Monitoring periods with capercaillie presence		
	1988–1993	1993–1998	1998–2003
'Continuously inhabited'	X	X	X
'Newly populated'	–	X	X
	–	–	X
'Abandoned'	X	–	–
	X	X	–
'Fluctuation'	X	–	X
	–	X	–

(X indicates capercaillie presence in the respective monitoring period, i.e., an area is classified 'continuously inhabited' if it was mapped as in all three monitoring periods).

Capercaillie habitat preferences with regard to forest vegetation and forest structure

(1) Tree species composition:	Coniferous and mixed Ideally dominated by Scots pine (*Pinus sylvestris*) but also Silver fir (*Abies alba*) or spruce (*Picea Abies*)
(2) Canopy cover:	Intermediate (50-70%)
(3) Forest age:	Old forest (> 70 years)
(4) Forest structure:	Well-structured mosaic of old forest interspersed with younger stages and gaps (e.g. small clearcuts (< 1 ha), or open areas following natural disturbance, e.g., windthrow, snow-break) Edges of bogs and mires High proportion of inner forest edges
(5) Ground vegetation:	High degree of ground vegetation cover Dominated by bilberry (*Vaccinium myrtillus*), but also northern bilberry (*Vaccinium uliginosum*), other ericaceous shrubs

(1) Scherzinger 1989, 1991; Klaus et al. 1989
(2) Gjerde 1991; Storch 1993c; Moss 1994; Suchant and Braunisch 2004
(3) Rolstad and Wegge1987, 1989; Storch 1993a,b,c
(4) Scherzinger 1989, 1991; Boag and Rolstad 1991; Suchant and Braunisch 2004
(5) Klaus et al. 1985; Picozzi et al. 1992; Storch 1993b; Sjöberg 1996; Selas 2000, 2001

Fig. 3. Capercaillie habitat preferences with regard to forest vegetation structures.

In a first step we separately evaluated the soil parameters HTY, STY, STX, NS and HR based on the knowledge of two site-mapping and vegetation experts not involved in capercaillie research. We assigned values between 0 and 1 according to the variable's potential to support suitable vegetation conditions for capercaillie (Fig. 3). These are open, well-structured forest types dominated by pine (*Pinus sylvestris*), but also European silver fir (*Abies alba*) and spruce (*Picea abies*), with a rich field layer dominated by *Vaccinium* species, preferably bilberry (*Vaccinium myrtillus*), but also cowberry (*Vaccinium vitis-idaea*), northern bilberry (*Vaccinium uliginosum*) and cranberry (*Vaccinium oxycoccus*). Such vegetation types are found on wet, raw humus dominated and acidic mineral soils and on organic material low in nutrients (e.g., Storm, 1996; Ihalainen and Pukkala, 2001; Härdtle et al., 2004), where slow growth rates, low increment and greater susceptibility to calamities (Braun et al., 2003; Mayer et al., 2005) also promote low forest stocking densities and high structural diversity. In order to integrate the stability of the soil status, we also included the regeneration potential of the different soil parameters with regard to past devastation. Sites characterised by particular soil types (e.g., dystric histosols), soil textures (e.g., extreme rocky sites) or combinations with particular nutrient or moisture regimes (e.g., very acidic wet mineral soils) exhibit no or only poor and slow regeneration potential (e.g., Glatzel, 1991; Buberl et al., 1994). Accordingly, soil parameters that characterised sites with a reduced nutrient turnover, with a slow reversal of historic acidification, and sites sensitive to the acidification of the topsoil were ranked higher than sites with a high regeneration potential. The values assigned to the different variables are provided in Table 2.

In a second step, the five soil parameters where weighted and combined to give a soil condition index (I_s), as shown in Eq. (1):

$$I_s = 5I_{HTY} + 5I_{STY} + 3I_{STX} + 2I_{NS} + \begin{cases} 1I_{HR}, & \text{if } I_{NS} > 0 \\ 0I_{HR}, & \text{if } I_{NS} = 0 \end{cases} \qquad (1)$$

where I_s is the soil condition index; I_{HTY}, I_{STY}, I_{STX}, I_{NS}, I_{HR} are the partial values of the soil parameters HTY, STY, STX, NS and HR as defined in Table 2 (ranging between 0 and 1).

2.4.2. Evaluation of the expert rating

In order to roughly estimate the validity of the parameter weighting and to obtain approximate reference values for the relative importance of the soil parameters, we performed a multiple linear regression, using the five soil parameters (I_{HTY}, I_{STY}, I_{STX}, I_{NS}, $I_{HR(if\ NS>0)}$) as independent variables and the proportion of bilberry cover as a response variable. The reason for choosing bilberry cover was that, of the assessed vegetation variables, bilberry was the most sensitive to soil conditions and the least actively influenced by forestry. Moreover, it is considered to be an indicator of capercaillie-relevant vegetation conditions (Schroth, 1994), it is correlated with all of the other vegetation structure variables given in Table 4 except ground vegetation cover (Suchant, 2002) and data were available for the entire study area. The data on bilberry cover were adopted from the national forest inventory and evaluated for 2984 sample plots (10 m radius) regularly distributed over the Black Forest study area (2 km × 2 km quadrangle grid; for details see: BMVEL, 2006). First, we compared the relative heights of the unstandardised regression coefficients of the soil parameters with the weights assigned by the experts. Subsequently, we compared the explanatory power of the multiple regression model with that of a regression using I_s as the only independent variable.

2.5. Correlation between soil condition and capercaillie-relevant vegetation structure variables (pilot area)

We tested the correlation between the soil condition index (I_s) and the current forest structure using the vegetation mappings in the pilot area. In addition, for each forest stand unit we combined the variables by calculating a habitat suitability index (HSI) for

Table 2

Results of the expert evaluation of the different soil condition parameters (A: humus type (HTY), B: soil type (STY), C: soil texture (STX), D: nutrient status (NS) and E: hydrological regime (HR)) with regard to their potential to support capercaillie-relevant vegetation types. The assigned partial values (I_{HTY}, I_{STY}, I_{STX}, I_{NS}, I_{HR}) range between 0 and 1. In addition, identification numbers (ID) are provided that allow the identification of the parameters from the site and soil condition database of Baden-Württemberg

(A) Humus type (HTY)		HTY_ID	Assigned value (I_{HTY})
Peat		60	1
Sphagnum (bog) peat		61	
Sphagnum (bog) peat, decomposed		62	
High moor peat, intact		63	
Peaty hydromor		54	0.6
Mor-like peat		66	
Hydromor		78	
Mormoder to mor		43	0.4
Mor		50	
Mor to moder		51	
Mor to mormoder		52	
Mor to hydromor		53, 55	
Hydromor to mor		76	
Degraded humus forms		80	
Mullmoder to mor		23	0.2
Mullmoder		24	
Mullmoder to mormoder		25	
Moder to dystrophic mor		35	
Moder to mor		36	
Moder to mormoder		40	
Mormoder		41	
Mormoder to moder		42	
Sedge peat		64	
Dystrophic mor		81	
Dystrophic mor to moder		82	
Other			0
(B) Soil type (STY)		**STY_ID**	**Assigned value (I_{STY})**
Dystric, fibric, folic, sapric, thionic	Histosol	920, 930	1
Stagnic	Planosol	620, 623	0.8
Histic	Planosol	621	
Dystric, spodic	Planosol	622	
Gleyic, alumic, dystric	Podzol	503, 504	0.6
Stagnic	Podzol	510, 615	
Stagni-lamellic, placic	Podzol	511	
Histic, mollic, umbric, humic	Gleysol	820	
Haplic	Podzol	500	0.4
Cambi-haplic	Podzol	501	
Leptic, lithic	Podzol	505	
Stagnic	Luvisol, albeluvisol	600	
Gleyi-stagnic	Luvisol	805	
Stagnic-dystric, stagni-eutric	Gleysol	610	
Spodic	Gleysol	801, 853	
Stagnic-vertic	Cambisol	613, 614	
Dystric, eutric	Planosol	612	
Spodic	Leptosol	132	0.2
Spodic	Cambisol	233, 234	
Haplic, eutric, dystric, umbric, mollic	Gleysol	802, 851	
Stagnic	Gleysol	804	
Dystric	Arenosol, cambisol	231, 232	0.1
Stagnic	Cambisol	240	
Other			0
(C) Soil texture (STX)		**STX_ID**	**Assigned value (I_{STX})**
Organic		94	1
Blocky		65	0.5
Sandy		40	0.2
Loamy		30, 50	0.1
Other			0
(D) Nutrient status (NS)		**NS_ID**	**Assigned value (I_{NS})**
Very acidic, dystric		25, 83	1
Devastated (due to grazing, litter raking) (relictic)		50, 52, 54	
Moderately poor		82	
Acidic		22	0.5
Other			0

Table 2 (*Continued*)

(E) Hydrological regime (HR)	HR_ID	Assigned value (I_{HR})
Very dry to dry	11	
Dry	12	
Moderately dry to dry	13, 14, 16	1
Dry to periodically dry	15	
Periodically dry	81	
Moderately dry	16	
Moist (ground water)	54, 55, 56	
Moist (ground water and stagnic water)	59	
Moist (ground water) to wet (spring water[a])	57	
Moist to wet (ground water)	58, 60	
Wet (ground water)	62	1
Wet to periodically moist (stagnic water)	61	
Periodically moist (stagnic water)	70, 71	
Periodically moist to wet (stagnic water)	72	
Wet (stagnic water)	73	
Periodically wet (stagnic water)	74	
Slightly moist to moist (ground water)	43	0.5
Slightly moist to periodically moist (stagnic water)	45	
Other		0

[a] Groundwater percolating to the surface through widely scattered vents.

capercaillie in winter, summer and for the whole year, using the model developed by Storch (2002). This model is based on the results of a telemetry study conducted in the Bavarian Alps (Storch, 1993a,b,c): In a first step, values between 0 and 1 are assigned to each of the variables, according to preferences or avoidance capercaillie exhibited in the respective season (winter or summer). In a second step, the variables are weighted and combined to a HSI for winter, summer and year. The vegetation variables included in the HSI_{winter} are forest type, successional stage and canopy cover. The HSI_{summer} includes successional stage, canopy cover, height of natural regeneration, bilberry cover and ground vegetation cover. The HSI_{year} is calculated as the geometric mean of the winter and the summer index (for details see Storch, 2002). We correlated each of the vegetation structure variables alone, and each of the HSI scores for summer, winter and year with I_s using Spearman's rank correlation (STATISTICA, StatSoft 1999).

2.6. Long- and short-term dynamics of capercaillie distribution in relation to soil condition (study area)

To quantify the relationship between soil condition and the spatial pattern of the long-term decline of the capercaillie distribution, we calculated the mean I_s within the areas (1, 10, 100, 500 and 1000 ha) surrounding historical lekking sites. These scales corresponded to the size of an average forest stand (10 ha), a small (100 ha) and a large (500 ha) capercaillie home range in the Black Forest (Braunisch and Thiel, unpublished data), and the mean size of an inhabited patch in the study area (1000 ha). We compared the categories '*still inhabited*' ($N = 55$) and '*abandoned*' ($N = 101$) using a Mann–Whitney U test. In addition, we employed logistic regression (Hosmer and Lemeshow, 1989) to predict the probability of a lekking site being either abandoned or still inhabited as function of the mean I_s.

The short-term changes were evaluated by comparing the mean I_s for areas of the categories '*continuously inhabited*' ($N = 55$), '*newly populated*' ($N = 37$) and '*abandoned*' ($N = 64$) using a Kruskal–Wallis test. To evaluate the effect of scale, we calculated the mean I_s within a circular moving window of 1, 10, 100, 500 and 1000 ha, and averaged the resulting I_s values over all grid cells for each area. Statistics were calculated using STATISTICA (StatSoft 1999).

3. Results

3.1. Soil condition index

3.1.1. Expert model

In the expert evaluation, 25 of 51 different humus types, 23 of 137 soil types, 4 of 11 soil textures, 5 of 27 classes of nutrient status and 18 of 48 different soil moisture regimes were assigned values > 0 (Table 2). Differences between the two expert estimates were rare, and divergences never exceeded an absolute value of 0.2. The agreement on the weighting and combination of the variables was similar. Soil condition index values in the study area ranged between 0 and 15 and were highest for extreme sites like mires and wet forests.

3.1.2. Evaluation of the expert rating

In the multiple regression model employed to estimate roughly the validity of the parameter weighting, all soil parameters (HTY, STY, STX, NS, HR) contributed significantly to explaining bilberry cover. The relative heights of the regression coefficients largely corresponded to the coefficients assigned to the respective soil parameter by the experts (Table 3), except for STX, which was

Table 3

Results of a multiple linear regression between the partial values of the single soil parameters (I_{HTY}, I_{STY}, I_{STX}, I_{NS}, $I_{HR(if\ NS>0)}$) and bilberry cover (%) assessed for 2984 sample plots, regularly distributed over the entire Black Forest study area. In addition to the standardised regression coefficients (BETA), the unstandardised regression coefficients (B) are provided, representing the multiplication factor of each soil parameter in the regression term. B-coefficients are also rescaled to values between 0 and 5 ($B_{(rescaled)}$) in order to compare them to the multiplication factors assigned to the parameters in the expert model (Factor I_s)

Variable	BETA	S.D. (BETA)	B	S.D. (B)	t(2978)	p-value	$B_{(rescaled)}$	(Factor I_s)
Constant			0,108	0,024	4,456	0,000		
I_{NS}	0,228	0,017	0,317	0,024	13,475	0,000	2,348	2,000
I_{HTY}	0,197	0,022	0,671	0,073	9,144	0,000	4,971	5,000
I_{STY}	0,160	0,020	0,675	0,084	8,049	0,000	5,000	5,000
I_{STX}	0,129	0,018	0,297	0,043	6,998	0,000	2,203	3,000
$I_{HR(if\ NS>0)}$	0,131	0,017	0,184	0,024	7,762	0,000	1,361	1,000

Table 4
Correlations between soil condition index and capercaillie-relevant vegetation structure variables and habitat suitability index values for winter, summer and the whole year

Variable category	Forest type	Forest structure			Ground vegetation		Habitat suitability (HSI)		
Variable	Conifer dominated forest	Proportion of old forest	Canopy cover	Gaps/open structures	Ground vegetation cover	Bilberry cover	HSI winter	HSI summer	HSI year
Unit	Number of sample sites	Number of sample sites	% Cover	Number of gaps/site	% Cover	% Cover	Index	Index	Index
Spearman's R	0.16	0.09	−0.10	0.07	0.20	0.22	0.14	0.19	0.18
$t(N-2)$	7.861	4.438	−4.939	3.620	9.838	10.941	6.987	9.642	9.153
p-value	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001

slightly overrated in the expert model. The total explanatory power of the multiple regression model based on the five soil condition parameters (multiple $R = 0.481$, $R^2 = 0.231$) did not differ significantly from the results of a model using the soil condition index as the only explanatory variable ($R = 0.449$, $R^2 = 0.201$) ($p > 0.05$, test for comparing correlation coefficients; STATISTICA, StatSoft, 1999).

3.2. Correlation between soil condition and capercaillie-relevant vegetation structure variables (pilot area)

We found soil conditions to be significantly correlated with all of the capercaillie-relevant forest vegetation structure variables investigated (Table 4), although the correlations were very low

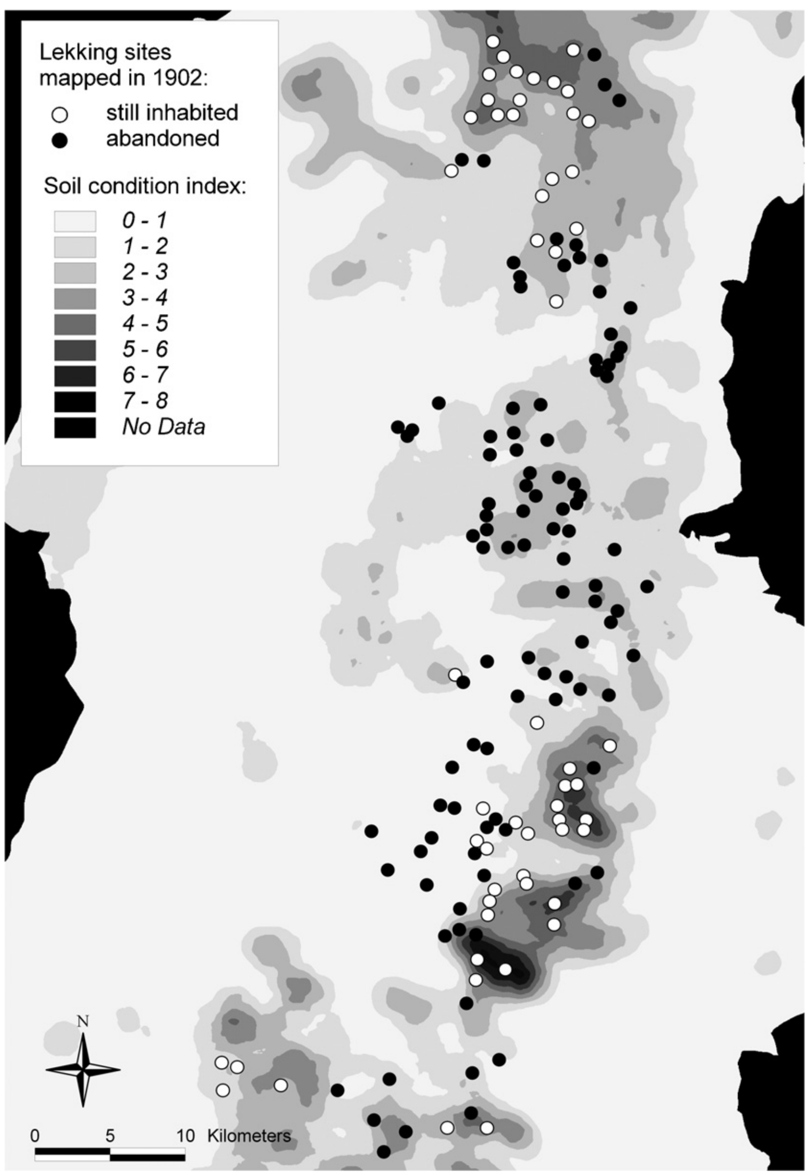

Fig. 4. Mean soil condition index within a circular moving window of 500 ha and the location of lekking sites mapped in 1902 – white dots: sites still inhabited; black dots: abandoned sites.

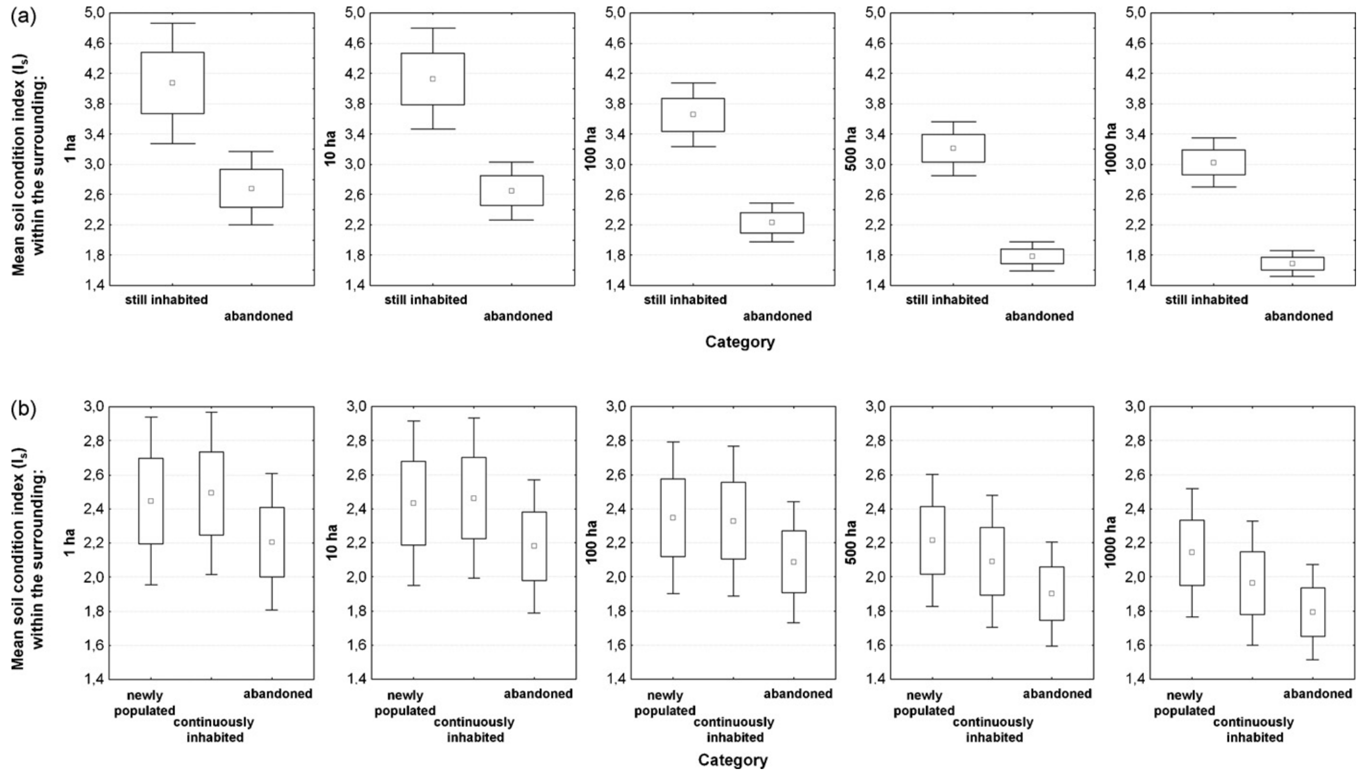

Fig. 5. Long- (a) and short-term (b) changes of capercaillie distribution in relation to soil condition at different spatial scales. Mean soil condition index within a surrounding area of 1, 10, 100, 500 or 1000 ha comparing (a) lekking sites (mapped in 1902) that were either 'still inhabited' or 'abandoned' within the last 100 years (1902–2003). (b) Areas 'newly populated', 'continuously inhabited' or 'abandoned' within the last 15 years (1988–2003). The central point indicates the mean, the box encompasses the standard error (S.E.) and the whiskers indicate 1.96 S.E.

and great variations were recorded between the forest stands. High soil condition indices related to vegetation characteristics that complied with the species preferences. The strongest correlation was recorded for ground vegetation, especially bilberry cover. This was not surprising, as the potential to support this key forage resource was a main criterion for the evaluation of the soil condition parameters in the expert model. Additionally, there was a higher number of conifer dominated forest types and a higher proportion of late successional stages. Sites with high soil condition indices were characterised by a lower canopy cover and a higher number of gaps, indicating greater structural diversity. Consequently, I_s was also positively correlated with habitat suitability indices for all seasons.

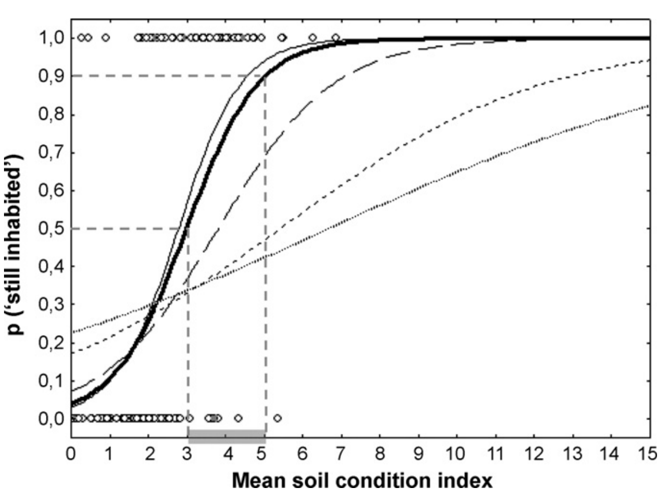

Fig. 6. Logistic regression models showing the probability that a lekking site of 1902 is still inhabited as a function of the mean soil condition index (I_s) within the surrounding areas of different sizes: 1 ha (dotted): $\chi^2 = 9.047$, $p < 0.01$; 10 ha (closely-spaced dashed): $\chi^2 = 15.110$, $p < 0.001$, 100 ha (widely-spaced dashed): $\chi^2 = 31.265$, $p < 0.0000$; 500 ha (bold continuous): $\chi^2 = 46.594$, $p < 0.0000$; 1000 ha (thin continuous): $\chi^2 = 49.808$, $p < 0.0000$). The data and the thresholds for predicting inhabitation with a probability of $p > 0.5$ and $p > 0.9$ respectively, are indicated for the 500 ha scale. The range of I_s crucial for habitat improvement measures for capercaillie is shaded grey.

3.3. Long- and short-term dynamics of capercaillie distribution in relation to soil condition (study area)

The pattern of the long-term contraction of the capercaillie range corresponded to the spatial pattern of suitable soil conditions (Fig. 4). The areas surrounding capercaillie lekking sites inhabited in 1902 but now abandoned revealed significantly lower soil condition indices than those areas still inhabited (Fig. 5a, Mann–Whitney U test: 1 ha scale: $U = 2089$, $p < 0.05$; 10 ha: $U = 1856$, $p < 0.001$; 100 ha: $U = 1365$, $p < 0.001$; 500 ha: $U = 1047$, $p < 0.001$; 1000 ha: $U = 1033$, $p < 0.001$). This applied to all spatial scales, but the differences became more pronounced as the scale increased.

The short-term development within the last 15 years revealed a similar trend. Abandoned sites showed slightly lower I_s than continuously inhabited patches at all spatial scales. Additionally, at the larger spatial scales (500 and 1000 ha), newly populated areas into which the distribution range had expanded in recent times revealed slightly higher I_s than continuously inhabited and abandoned patches. This was not statistically significant, however (Fig. 5b, Kruskal–Wallis test: $p > 0.05$ for all spatial scales).

The probability of a historical lekking place being 'still inhabited' increased logistically with the mean soil condition index in the surrounding area. Again, the model performance improved with increasing spatial scale (Fig. 6) revealing almost similar results for the 500 and 1000 ha scale (Fig. 6). At the 500 ha scale a mean soil condition index of 3.0 was recorded as the threshold for predicting inhabitation with a probability of more than 50% ($p_{(\mathrm{inhabitation})} > 0.5$). From I_s 3–5 the probability of long-term inhabitation increased steeply and reached the plateau for $I_s > 6$. The respective threshold values were slightly lower ($I_s = 2.8$ for $p_{(\mathrm{inhabitation})} > 0.5$) at the 1000 ha scale.

4. Discussion

4.1. Soil condition index

The aim of this study was to evaluate the potential use of forest site and soil mapping data to predict areas where the soil conditions naturally support vegetation conditions suitable for capercaillie. In managed forests, the soil-dependent development potential of vegetation structure does not necessarily coincide with the current soil–vegetation relationship. Therefore, we preferred an expert model over a statistical model to quantify the importance of the different soil parameters. However, expert models are quite sensitive to uncertainties, particularly differing expert opinions (e.g., Roloff and Kernohan, 1999; Johnson and Gillingham, 2004). In this study, both model steps, the rating of the single soil parameters (Table 2) and their weighting and combination (Eq. (1)), were based on the estimates of two vegetation experts. Although their ratings largely agreed, possible errors cannot be ruled out completely. While the results for the first step were not evaluated separately, we obtained an approximate estimate of the parameter weights by comparing the expert weightings with the unstandardised regression coefficients of a multiple regression. This was technically possible as the index was based on a linear combination of the soil parameters, and variable interactions were only considered between nutrient status and hydrological regime. The results indicated that the experts' weightings were valid, at least for predicting bilberry cover. They also showed that the low correlations between I_s and bilberry cover (Table 4) were not due to an incorrect parameter weighting, as the overall correlation and the amount of explained variance of the multiple regression model did not differ to a model using only I_s as independent variable. However, this evaluation can only provide a rough estimate of the relative parameter weights for two reasons. First, although bilberry cover was shown to be a surrogate for the vegetation status suitable for capercaillie in the Black Forest (Schroth, 1994; Suchant, 2002) the relative importance of the soil parameters with regard to the other vegetation variables may not be considered sufficiently. Second, the analysis referred to the current vegetation status (bilberry cover), which may be altered locally by forest management. Consequently, the evaluation may be of limited suitability for evaluating an index designed to describe an 'ideal', i.e., unbiased interaction. Although the results indicate a sufficient model accuracy, the soil condition index might be further improved by including the estimates of a larger number of experts, and by quantifying the impact of possible estimate errors using a sensitivity analysis.

4.2. Soil conditions and vegetation structure

Although significant, the correlations between I_s and the investigated vegetation structure variables were very low, mainly due to a high variance recorded over the forest stands. In addition to the fact that small scale vegetation structures inherently vary considerably in dependence of natural succession processes, the soil–vegetation interaction in the study area is largely influenced by forest management which continuously alters and determines forest structure and composition. However, depending on the prevailing soil conditions, the same management regime applied to different sites may lead to contrasting results, e.g., due to different growth rates but also different susceptibilities to snow damage and insect calamities (Braun et al., 2003; Mayer et al., 2005). These purely soil-dependent differences in vegetation structures, reflected by the correlations obtained, are expected to be small compared to the total variance. Moreover, the soil–vegetation interaction was only analysed for a small part of the study area, where soil conditions were rather homogeneous (Fig. 2). Stronger correlations may be expected when considering the total study area, covering a wider range of soil conditions. This assumption is supported by the correlation between I_s and bilberry cover for the whole study area, which was stronger than for the pilot area alone. Nevertheless, even in intensively managed forests significant correlations between soil condition and both vegetation structure and capercaillie habitat suitability were obtained.

4.3. Soil conditions and capercaillie distribution

The relevance of soil condition as an indirect predictor of capercaillie habitat selection has received little attention to date. Graf et al. (2005) integrated extreme sites (mires and wet forests) into their habitat model. These corresponded to the sites with the highest soil condition indices ($I_s > 9$) in this study, which for the first time provides a differentiated evaluation of soil- and site-related variables. We found soil condition to be an important indirect predictor of long-term capercaillie presence, performing best when referring to a larger surrounding area of 500–1000 ha. This result matches well with those of Graf et al. (2005), who found that the proportion of mires within an area of 529 ha best explained capercaillie presence in the Swiss Alps. Although we demonstrate that high mean soil condition indices were most favourable within the area of a large home range up to the mean size of an inhabited forest patch, we could not specify whether the value of a small number of interspersed sites with a high I_s (like small mires or wet forests) was equal to an intermediate I_s throughout the whole area. This aspect, which may be crucial to successful habitat management, requires further investigation.

The contraction of the capercaillie range to sites with high I_s is most obvious when considering the significant differences between abandoned lekking sites and those sites still inhabited since 1902 (Fig. 5a). The short-term trend, showing that sites abandoned over the last 15 years tended to exhibit a lower I_s than average, whereas newly occupied areas had a slightly higher value, indicates that this process is still ongoing (Fig. 5b). These differences were not significant, but a period of 15 years is a comparatively short timeframe for the evaluation of changes to the distribution range of this bird species, characterised by a long lifespan (up to 10 years in the wild, Cramp and Simmons, 1980) and a high degree of site-fidelity.

As no large-scale, spatially explicit historic data on vegetation structure and soil conditions were available for the capercaillie distribution area, it was not possible to relate the change of the capercaillie distribution to the changes of vegetation structure, and to measure the change rates for different soil conditions, which may have supported a causality in the observed relationship. However, the combination of both results, the significantly more frequent abandonment of sites with low I_s and the correlations between I_s and habitat suitability corroborates a likely effect of soil condition on long-term capercaillie presence.

The comparison of inhabited and abandoned sites (Fig. 5) provides a supplementary hint at the link between soil condition and capercaillie habitat suitability. The areas surrounding still inhabited lekking sites (Fig. 5a) always exhibited higher mean I_s than the continuously inhabited and newly populated areas used for evaluating the short-term trends in capercaillie inhabitation (Fig. 5b). This is reasonable, considering that the former areas are restricted to the vicinity of lekking sites, which usually exhibit the best habitat quality (e.g., Wegge and Rolstad, 1986; Storch, 1997), whereas the latter areas also include other parts of the distribution range.

4.4. Application in species conservation and habitat management

Differentiating between sites where favourable habitat conditions depend on repeated management actions and alternatively, sites where the preferred vegetation status is likely to remain suitable with either no or only very little management effort, can be an important cue for decision making in species conservation and wildlife management. A practical consequence for capercaillie conservation would be to concentrate habitat improvement measures (like the reduction of stocking densities to further the proliferation of bilberry and increase structural diversity) on sites with high soil condition indices, especially where forestry has created unsuitable, homogenous forest structures. Taking an advantage of the natural dynamics of such sites would lead not only to an ecological but also to an economic optimisation of conservation efforts. The I_s values at which long-term capercaillie inhabitation becomes likely (e.g., $p_{(\mathrm{inhabitation})} > 0.5$, Fig. 6), considered within an area that meets the spatial requirements of the bird (>500 ha), can serve as a threshold for management implications. At sites with $I_s > 5$, for example, there is a high probability of capercaillie inhabitation ($p_{(\mathrm{inhabitation})} > 0.9$) without or with only very little additional support. Given that the total area and spatial configuration of these sites ($I_s > 5$) in a region is not sufficient to maintain a viable population, the best cost-benefit ratio can be achieved by concentrating active and recurrent habitat improvement measures on sites where I_s is still high enough to support management objectives and where an additional input can increase considerably the probability of inhabitation (e.g., I_s 3–5, Fig. 6). At sites with larger I_s values, where soil conditions are optimal but vegetation structures have been degraded, single restoration measures are most promising. Subsequent to the large-scale prioritisation of conservation relevant sites, the soil maps can also support the planning of the concrete measures at a local scale (Fig. 2). The fact that area-wide soil maps are available, and that the mapping units serve as a foundation for forest planning, facilitates an integration of conservation measures into regular forest practice.

4.5. Conclusions

Ecosystems change continuously as a result of natural processes and human land use. The dynamics of these changes largely depend on the given landscape conditions, such as site and soil. This study shows that soil conditions can be an important indirect predictor of habitat suitability for wildlife species highly specialised to particular vegetation types and structures. In regions where human land use practices have greatly altered the vegetation structure, information about soil condition can help to predict long-term trends in changes of vegetation, habitat suitability and species distribution ranges. Forest site and soil inventories provide area-wide, spatially explicit data that can support the prioritisation of sites for habitat restoration measures. As the individual site and soil condition parameters are mapped independently of a current or projected use, they may be combined and evaluated according to the respective requirements of different problems and target species.

However, soil condition represents only one of the factors that determine the potential for vegetation-related habitat suitability. Especially in mountainous landscapes, climate is a predominant factor, not only with regard to vegetation development but also to the regeneration potential of devastated soils (Prietzel and Rehfuess, 1999). Climate conditions also count among the most important predictors for capercaillie presence at the landscape scale (e.g., Sachot, 2002; Graf et al., 2005), with cold winter conditions best explaining capercaillie distribution in the Black Forest (Braunisch and Suchant, 2007). To obtain more precise predictions, climate and topography must be additionally included, and the interactions between these variables quantified. Furthermore, predicting future developments from the current status also requires the inclusion of long-term changes, for example, climate change and atmospheric nitrogen deposition.

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Erratum

Erratum to "Using ecological forest site mapping for long-term habitat suitability assessments in wildlife conservation—Demonstrated for capercaillie (Tetrao urogallus)" [Forest Ecol. Manag. 256 (2008) 1209–1221]

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Due to an unfortunate error in the publication process Table 2 was printed in error. The correct table is given below.

Table 2
Results of the expert evaluation of the different soil condition parameters (a: humus type (HTY), b: soil type (STY), c: soil texture (STX), d: nutrient status (NS) and e: hydrological regime (HR)) with regard to their potential to support capercaillie-relevant vegetation types. The assigned partial values (I_{HTY}, I_{STY}, I_{STX}, I_{NS}, I_{HR}) range between 0 and 1. In addition, identification numbers (ID) are provided that allow the identification of the parameters from the site and soil condition database of Baden-Württemberg

(a) Humus type (HTY)	HTY_ID	Assigned value (I_{HTY})
Peat	60	1
Sphagnum (bog) peat	61	
Sphagnum (bog) peat, decomposed	62	
High moor peat, intact	63	
Peaty hydromor	54	0.6
Mor-like peat	66	
Hydromor	78	
Mormoder to mor	43	0.4
Mor	50	
Mor to moder	51	
Mor to mormoder	52	
Mor to hydromor	53, 55	
Hydromor to mor	76	
Degraded humus forms	80	
Mullmoder to mor	23	0.2
Mullmoder	24	
Mullmoder to mormoder	25	
Moder to dystrophic mor	35	
Moder to mor	36	
Moder to mormoder	40	
Mormoder	41	
Mormoder to moder	42	
Sedge peat	64	
Dystrophic mor	81	
Dystrophic mor to moder	82	
Other		0

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Table 2 (*Continued*)

(b) Soil type (STY)		STY_ID	Assigned value (I_{STY})
Dystric, fibric, folic, sapric, thionic	Histosol	920, 930	1
Stagnic	Planosol	620, 623	0.8
Histic	Planosol	621	
Dystric, spodic	Planosol	622	
Gleyic, alumic, dystric	Podzol	503, 504	0.6
Stagnic	Podzol	510, 615	
Stagni-lamellic, placic	Podzol	511	
Histic, mollic, umbric, humic	Gleysol	820	
Haplic	Podzol	500	0.4
Cambi-haplic	Podzol	501	
Leptic, lithic	Podzol	505	
Stagnic	Luvisol, albeluvisol	600	
Gleyi-stagnic	Luvisol	805	
Stagnic-dystric, stagni-eutric	Gleysol	610	
Spodic	Gleysol	801, 853	
Stagnic-vertic	Cambisol	613, 614	
Dystric, eutric	Planosol	612	
Spodic	Leptosol	132	0.2
Spodic	Cambisol	233, 234	
Haplic, eutric, dystric, umbric, mollic	Gleysol	802, 851	
Stagnic	Gleysol	804	
Dystric	Arenosol, cambisol	231, 232	0.1
Stagnic	Cambisol	240	
Other			0
(c) Soil texture (STX)		STX_ID	Assigned value (I_{STX})
Organic		94	1
Blocky		65	0.5
Sandy		40	0.2
Loamy		30, 50	0.1
Other			0
(d) Nutrient status (NS)		NS_ID	Assigned value (I_{NS})
Very acidic, dystric		25, 83	1
Devastated (due to grazing, litter raking) (relictic)		50, 52, 54	
Moderately poor		82	
Acidic		22	0.5
Other		0	
(e) Hydrological regime (HR)		HR_ID	Assigned value (I_{HR})
Very dry to dry		11	1
Dry		12	
Moderately dry to dry		13, 14, 16	
Dry to periodically dry		15	
Periodically dry		81	
Moderately dry		16	
Moist (ground water)		54, 55, 56	1
Moist (ground water and stagnic water)		59	
Moist (ground water) to wet (spring water[a])		57	
Moist to wet (ground water)		58, 60	
Wet (ground water)		62	
Wet to periodically moist (stagnic water)		61	
Periodically moist (stagnic water)		70, 71	
Periodically moist to wet (stagnic water)		72	
Wet (stagnic water)		73	
Periodically wet (stagnic water)		74	
Slightly moist to moist (ground water)		43	0.5
Slightly moist to periodically moist (stagnic water)		45	
Other		0	

[a] Groundwater percolating to the surface through widely scattered vents.